

LOEI, Thailand

WMO: 483530

Lat: 17.4528N

Lon: 101.7306E

Elev: 254

StdP: 98.31

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.8	13.7	7.3	6.5	19.3	9.2	7.5	19.8	4.4	25.8	4.1	26.0	0.3	0	0.282

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.9	38.1	23.4	36.8	23.5	35.4	23.8	26.7	31.8	26.4	31.4	26.1	31.1	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	21.5	28.9	25.2	20.9	28.4	24.9	20.6	28.1	86.2	32.0	84.3	31.7	82.9	31.5	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.0	4.2	3.5	DB	8.8	39.8	2.6	1.4	6.9	40.8	5.4	41.6	3.9	42.4	2.0	43.4	(4)
(5)			WB	7.9	28.1	2.4	1.0	6.2	28.8	4.8	29.4	3.4	29.9	1.7	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	22.8	24.9	27.4	29.2	28.7	28.7	28.2	27.8	27.4	26.7	25.0	22.6
	DBStd	2.91	2.42	2.65	2.36	2.03	1.72	1.25	1.27	1.21	1.14	1.44	2.10	2.68
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	7	2	1	1	0	0	0	0	0	0	0	0	3
	CDD10.0	6072	398	418	539	576	581	561	566	553	523	517	450	391
	CDD18.3	3037	142	186	282	326	322	311	307	294	273	259	200	136
	CDH23.3	30182	1236	1907	3165	3774	3374	3364	3187	2853	2423	2208	1622	1069
CDH26.7	12793	458	943	1739	2041	1496	1379	1197	988	807	805	604	336	
Wind	WSAvg	1.1	1.0	1.1	1.2	1.2	1.1	1.1	1.2	1.2	1.0	0.9	0.9	0.9
Precipitation	PrecAvg	1218	6	14	38	93	203	148	137	189	249	111	17	4
	PrecMax	1543	43	53	67	178	389	339	196	351	528	211	54	40
	PrecMin	1008	0	0	1	20	78	49	52	103	131	20	1	0
	PrecStd	155	13	17	20	48	81	74	43	74	116	61	18	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	36.9	39.1	40.6	38.1	36.0	35.6	34.3	34.0	34.0	34.2	33.0
		MCWB	21.0	21.8	22.3	23.6	25.3	25.6	25.5	25.6	25.9	25.3	23.8	22.9
	2%	DB	32.2	35.3	37.7	38.9	36.1	34.5	33.9	33.1	32.9	32.7	32.7	31.2
		MCWB	21.3	21.5	22.1	24.0	25.4	25.5	25.3	25.5	25.7	24.9	23.7	22.3
	5%	DB	30.9	34.0	36.3	37.1	34.6	33.4	32.8	32.1	31.9	31.8	31.4	30.0
		MCWB	21.0	21.7	22.3	24.2	25.6	25.4	25.2	25.3	25.5	24.5	23.2	21.7
	10%	DB	29.4	32.4	34.8	35.4	33.1	32.3	31.7	31.2	30.9	30.8	30.2	28.7
		MCWB	20.4	21.6	22.6	24.3	25.5	25.4	25.1	25.1	25.3	24.3	22.7	20.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.5	25.2	26.4	27.3	27.1	26.9	27.0	27.1	26.6	25.6	24.8
		MCDB	30.1	32.7	32.9	34.5	33.4	32.1	31.8	31.4	31.4	31.1	30.7	29.8
	2%	WB	23.0	23.7	24.5	25.7	26.6	26.5	26.2	26.4	26.4	26.0	24.9	23.7
		MCDB	28.8	31.6	32.6	33.5	32.6	31.6	31.0	30.8	30.7	30.4	29.7	28.6
	5%	WB	22.1	23.0	23.9	25.3	26.1	26.2	25.9	26.0	26.0	25.5	24.3	22.7
		MCDB	27.9	30.7	32.2	32.8	32.0	31.1	30.5	30.3	30.3	29.8	28.8	27.6
	10%	WB	21.3	22.3	23.5	24.9	25.7	25.8	25.5	25.6	25.6	25.0	23.7	22.0
		MCDB	27.3	29.5	31.3	32.2	31.1	30.5	29.8	29.6	29.6	29.1	28.1	27.1
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.6	13.8	11.9	8.9	7.7	7.2	6.9	7.2	8.6	10.7	12.3
		MCDBR	15.6	16.9	17.1	15.2	11.5	9.4	8.8	8.4	8.8	9.9	12.0	13.0
		MCWBR	5.8	5.4	4.7	3.5	3.0	2.5	2.5	2.7	3.0	3.3	4.1	4.9
	5% WB	MCDBR	12.9	14.4	13.6	12.1	9.7	8.2	7.7	7.7	7.9	8.8	9.9	11.4
		MCWBR	5.0	4.9	4.2	3.5	3.1	2.7	2.7	2.8	3.1	3.3	3.7	4.5
Clear-Sky Solar Irradiance	taub	0.466	0.573	0.674	0.626	0.495	0.445	0.433	0.448	0.495	0.540	0.454	0.456	
	taud	2.108	1.848	1.653	1.801	2.188	2.356	2.385	2.321	2.174	2.006	2.221	2.155	
	Ebn at Noon	812	743	685	722	813	846	859	852	813	761	820	809	
	Edh at Noon	151	206	258	223	150	125	122	131	151	173	134	141	
All-Sky Solar Radiation	RadAvg	4.65	5.09	5.14	5.51	5.18	4.86	4.55	4.41	4.50	4.70	4.83	4.50	
	RadStd	0.31	0.23	0.33	0.30	0.36	0.35	0.31	0.28	0.28	0.32	0.20	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	+4	+130	+128
(47)	Regional (17 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+116

Nomenclature: See separate page